ALEXIS at DialectSentEval Subtask 1: Adversarial Training and Iterative Self-Training for Zero-Shot Dialect Sentiment Analysis

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Abstract

Dialect sentiment analysis asks for the polarity of a sentence written in a regional Arabic variety rather than Modern Standard Arabic. We present a system for Subtask 1 of DialectSentEval built on two observations about the data. First, the task is one of zero-shot transfer rather than classification: the test set introduces Lebanese, a dialect with no labeled training data and the highest out-of-vocabulary rate in the task (52.0% of word types, against 19.5–41.0% for the seen dialects). Second, k-fold cross-validation cannot see this, since it measures generalization to unseen sentences and never to an unseen dialect. We therefore validate with a leave-one-dialect-out diagnostic and score every component on the official leaderboard. Three dialectally diverse encoders (MARBERTv2, CAMeLBERT-DA, AraBERTv2) are trained with Fast Gradient Method adversarial training (Miyato et al., 2017), then refined by confidence-filtered self-training on the unlabeled test set and character-level noise augmentation (Srivastava and Chiang, 2023). Adversarial training contributes +0.040 Macro F1 over a gold-only ensemble, self-training a further +0.027, and per-class pseudo-label filtering +0.018. The final system reaches 0.8667 Macro F1, ranking **1st** of 12 submissions, 4.6 points above the runner-up.

1 Introduction

Sentiment analysis is a mature task in Arabic NLP, but most user-generated Arabic text is dialectal rather than Modern Standard Arabic (MSA). Dialects differ enough in vocabulary and orthography that a model tuned on one can fail on another, which makes cross-dialect transfer the interesting part of the problem.

Subtask 1 of DialectSentEval (Ezzini et al., 2026a), hosted with the Arabic Natural Language Processing Conference (ArabicNLP 2026),[1] asks participants to classify hotel-review sentences from the MDS-3 dataset (Alharbi et al., 2025c) into {positive, negative, neutral} for Moroccan (Darija), Egyptian, Jordanian, and Saudi. The released test set adds a fifth dialect, Lebanese, with no labeled training data at all. The subtask continues the AraSentEval (Ezzini et al., 2026b) and AHASIS (Alharbi et al., 2025b) shared tasks, and our design draws on prior benchmarking of fine-tuned encoders and large language models on closely related Arabic dialect sentiment data (Alharbi et al., 2025a). Treating an unseen dialect as its own generalization regime follows prior work on quantifying dialectal distance (Sakr et al., 2024) and on identifying a dialect from only a handful of labeled examples (Yusuf et al., 2022).

Hypothesis. Our starting point is that standard in-distribution validation is the wrong instrument for this task. k-fold cross-validation measures generalization to unseen *sentences* drawn from dialects the model has already seen; it says nothing about generalization to an unseen *dialect*. Our exploratory analysis confirms that Lebanese is exactly the out-of-distribution case: it has the highest word-type out-of-vocabulary rate of any test dialect (52.0%, against 19.5–41.0% for the four seen dialects), and, as a Levantine variety, it is linguistically closest to Jordanian.

That diagnosis motivates two decisions, each aimed at an identified failure mode rather than adopted as a generic best practice:

- A **leave-one-dialect-out (LODO)** protocol that simulates the real unseen-dialect condition during development.
- **Embedding- and character-level perturbation** (FGM and character noise) as explicit proxies for the lexical and orthographic distance between a seen dialect's surface forms and an unseen one's.

[1] https://ezzini.github.io/DialectSentEval2026/. Our code: GitHub.

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We tested the hypothesis with a controlled ablation across five axes: pooling, adversarial and consistency regularization, domain invariance, character-aware architectures, and self-training. Each variant was scored independently on the leaderboard against a common single-backbone baseline. As Section 5 shows, the interventions that match the diagnosed failure mode survived; the generic regularizers and domain-invariance objectives did not.

2 Data and Exploratory Analysis

MDS-3 (Alharbi et al., 2025c) contains 1,731 labeled training rows and 525 unlabeled test rows of hotel-review sentences, each translated into every dialect from a shared underlying set of reviews. Table 1 gives the split.

Dialect	Train	Test
Darija (Moroccan)	433	105
Egyptian	432	105
Jordanian	433	105
Saudi	433	105
Lebanese	0	105
Total	**1,731**	**525**

Table 1: Rows per dialect. Lebanese has zero training examples.

Labels are not a confound. The three classes are close to balanced (negative 38.0%, positive 35.2%, neutral 26.9%) and near-uniform across dialects. Any dialect-wise performance gap therefore reflects genuine transfer difficulty rather than a shifting label distribution.

Duplication. Seventeen training sentences are exact duplicates of one another, and 58 test sentences are byte-identical to a training sentence. Both facts shape the system: the first determines how we group cross-validation folds, the second motivates the exact-match lookup described in Section 3.

Vocabulary overlap. Figure 1 plots the fraction of whole-word token *occurrences* in each test dialect that are absent from the training vocabulary. On this measure Lebanese reaches 34.3%, more than 1.5× the rate of any seen dialect; measured over token *types* instead, the gap is wider still (52.0%, quoted above). Either way the conclusion is the same: a model that leans on memorizing dialect-specific function words will fail on Lebanese in particular. This is what pushed us toward dialectally pretrained backbones (Section 3) and toward LODO rather than random k-fold validation alone.

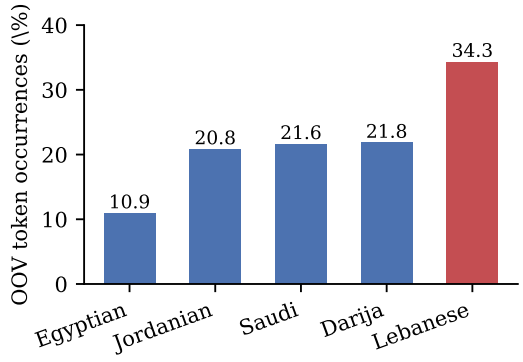


Figure 1: Out-of-vocabulary token-occurrence rate by test dialect. Lebanese (highlighted), absent from training, has the highest rate.

Negation and length. Negation markers appear in 44.4% of negative sentences but also in 37.8% of neutral ones, so surface cues alone separate the classes poorly and a contextual transformer is preferable to a lexicon- or rule-based classifier. Sentences are short to medium (median 16–19 words), so capping input at 128 sub-words discards almost nothing.

3 System Overview

Backbones. We fine-tune three encoders, chosen for dialectal-Arabic coverage and for diversity from one another. `UBC-NLP/MARBERTv2` (Abdul-Mageed et al., 2021) was the strongest single model on the AHASIS pilot of a related dataset (Alharbi et al., 2025b). `CAMeL-Lab/bert-base-arabic-camelbert-da` (Inoue et al., 2021) is pretrained on Dialectal Arabic and is the organizers' official sentiment classifier for Subtask 2. `aubmindlab/bert-base-arabertv2` (Antoun et al., 2020) is MSA-centric and is included precisely because it errs differently from the other two.

Pooling and loss. Each backbone uses masked mean pooling rather than the `[CLS]` token, so that a handful of unfamiliar Lebanese sub-words cannot dominate a single aggregation point. Training uses class-weighted cross-entropy to offset the mild under-representation of the neutral class (26.9% of

rows), which matters because Macro F1 penalizes minority-class errors heavily.

Adversarial training. Every backbone is trained with Fast Gradient Method (FGM) adversarial training (Miyato et al., 2017). At each step a bounded perturbation is added to the word-embedding layer along the direction of the loss gradient, and the model backpropagates through the perturbed forward pass as well as the clean one. This is the single largest individual gain in our search (Section 5). We read it as a proxy for the robustness a model needs when moving from a seen dialect's embedding neighborhood to an unseen one's.

Iterative self-training. We exploit the unlabeled test set directly, over three rounds. In each round a committee of the current best models labels the test set; rows on which the committee agrees above a confidence threshold enter the training pool as pseudo-labels with a reduced loss weight ($0.7\times$), and a fresh model is trained on gold plus pseudo-labeled data. Round 3 adds a mild per-class confidence filter that trims the least-confident 10% of each class's own pseudo-labels. The per-class form matters: a flat threshold systematically underrepresents the neutral class, whose predictions are less confident than positive or negative ones. Coverage grew from 386 of 525 rows in round 1 to 499 by round 3.

Character-level noise augmentation. Following Srivastava and Chiang (2023), who show that character noise injected during fine-tuning improves zero-shot transfer to unseen dialects, we add one noised duplicate of every gold row. The noise combines Arabic letter-variant substitutions (ة/ه, ى/ي), which reflect real orthographic variation, with generic delete, duplicate, and swap operations at roughly 6% per character. This attacks the Lebanese OOV gap directly and makes no assumption about Lebanese class proportions.

Ensembling and dataset-overlap control. The three backbones are combined by uniform averaging of softmax probabilities. Our exploratory analysis also surfaced the 58 test sentences (11.0%) that are byte-identical to training sentences, an artifact of how the shared task's data was constructed. Left alone, these rows conflate memorization with generalization. We treat them as a data-quality issue rather than a scoring opportunity: an exact-match lookup returns the known gold label for them, so the model predictions and the ablation comparisons in Section 5 are evaluated on the 89.0% of rows that genuinely require generalization.

4 Experimental Setup

All models are fine-tuned on the full 1,731 labeled rows, plus self-training pseudo-labels where applicable. Development used two validation protocols side by side. The first is five-fold Stratified-GroupKFold, grouped by exact sentence text so that the 17 duplicated training sentences never straddle a fold boundary. The second is leave-one-dialect-out, holding out each seen dialect in turn; only LODO approximates the test set's unseen-Lebanese condition, so it served as our primary diagnostic. Table 2 lists the final recipe's key hyperparameters.

Hyperparameter	Value
Max sequence length	128 sub-words
Pooling	Masked mean
Loss	Class-weighted CE
Class weights	sklearn 'balanced'
(neg./pos./neu., per-fold)	≈0.88 / 0.95 / 1.24
FGM perturbation norm (ϵ)	1.0
Self-training rounds	3
Committee confidence threshold	0.55
Pseudo-label loss weight	$0.7\times$
Char-noise rate (per character)	≈6%
Ensemble combination	Uniform softmax avg.

Table 2: Key hyperparameters of the final deployed system.

5 Results

Table 3 reproduces the official leaderboard. Our final submission reached Macro F1 = 0.8667 and Accuracy = 0.8857, ranking first of twelve teams and 4.6 points ahead of the runner-up (0.8210).

#	Team/Participant	Macro F1
1	**ALEXIS (momen_ibrahim)-870387**	**0.8667**
2	ajwad_hossain-867052	0.8210
3	tasneemduridi-857223	0.8174
4	habiba_ayman-867381	0.8140
5	normalman159-857755	0.8010
6	ahmadalmoustafa-856400	0.7977
7	mona_-871069	0.7953
8	faisalm93-869067	0.7939
9	hoanguitcs-866914	0.7936
10	shadenalzeer-867546	0.7894
11	hh03112005-869469	0.7859
12	23020073-870754	0.7722

Table 3: Official Subtask 1 leaderboard: Arabic dialect sentiment analysis.

5.1 Ablation

Every row of Table 4 is an independently leaderboard-scored configuration rather than an internal estimate. The upper block traces the deployed system through its four milestones; each hypothesis-motivated component earns its place.

Configuration	F1
Baseline (gold-only ensemble)	0.7806
+ FGM adversarial (single backbone)	0.8204
+ Self-training, 3 rounds	0.8477
+ Mild per-class pseudo-label filter	0.8656
+ Char-noise augmentation (final)	**0.8667**
Other regularizers[a]	0.785–0.807
Fish (Shi et al., 2021)	0.3215
ByT5 backbone (5-fold CV)	0.376

Table 4: Top: cumulative deployed-system ablation, each row independently leaderboard-scored. Bottom: alternative single-backbone techniques tried and not deployed, none surpassing FGM's 0.8204. (ByT5 (Xue et al., 2022) was never submitted; its CV number is not comparable to the official F1.) [a]R-Drop (Liang et al., 2021), SWAD (Cha et al., 2021), Invariant Risk Minimization (Arjovsky et al., 2019), CORAL (Sun and Saenko, 2016), Supervised Contrastive Loss (Khosla et al., 2020), MixStyle (Zhou et al., 2021), and character CNN/BiLSTM hybrids.

The most instructive result is a failure, and it is a methodological one. An earlier, more aggressive pseudo-label filter forced class proportions to match those of the four *seen* dialects exactly, and it regressed on the real leaderboard (0.8336 against 0.8477). It failed for the same reason the whole task is hard: it assumed Lebanese shares the seen dialects' distribution. The milder, dialect-agnostic filter makes no such assumption, and it is what we deployed.

The pattern in the lower block is consistent with our hypothesis. Every non-surviving technique targets generic regularization or broad domain invariance rather than the specific OOV mechanism identified in Section 2. Fish and ByT5 did not merely underperform; they failed outright.

5.2 Error Analysis

Table 5 breaks LODO performance down by held-out dialect and class. Jordanian, Lebanese's Levantine relative, is the easiest to recover zero-shot (F1 = 0.980) and is our closest available proxy, though an optimistic one: LODO still draws held-out examples from the same translated-review pipeline as training, whereas Lebanese is genuinely unseen. Egyptian is the hardest held-out case (F1 = 0.899) despite no particular linguistic distance from the rest, so dialect family alone does not fully predict transfer difficulty. Neutral is the hardest class to recover across all four splits.

Held-out	Neg. F1	Neu. F1	Pos. F1
Jordanian	0.988	0.970	0.983
Saudi	0.976	0.950	0.984
Darija	0.911	0.888	0.933
Egyptian	0.928	0.856	0.914

Table 5: LODO per-class F1 when each seen dialect is held out. Neutral is consistently the hardest class to recover zero-shot.

6 Conclusion

Team ALEXIS's Subtask 1 submission ranked first of twelve teams with Macro F1 = 0.8667, by diagnosing one failure mode, zero-shot transfer to an unseen dialect, choosing interventions that match it, and validating throughout with a LODO protocol ordinary cross-validation cannot provide.

The gap between seen-dialect performance (Macro F1 above 0.93 in cross-validation) and the official score confirms Lebanese remains the dominant error source even for the best system. Closing it further will likely require genuine Lebanese-domain signal rather than more regularization on the dialects we already have.

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